NSF

ACCESS

Support

NAIRR Pilot

National Artificial Intelligence Research Resource Pilot

University of Colorado Boulder

Introduction to AI-Assisted Coding: Prompt Engineering with LLMs in Google Colab

Burch Fisher and Xiaoxu Guo
Associate Research Scientists/NSF SCIPE Cyberinfrastructure Professionals
Chesapeake Global Collaboratory
University of Maryland Center for Environmental Science

Supported by NSF grant #2321008

University of Maryland
CENTER FOR ENVIRONMENTAL SCIENCE

Research Computing
UNIVERSITY OF COLORADO BOULDER

NSF

ACCESS
Support

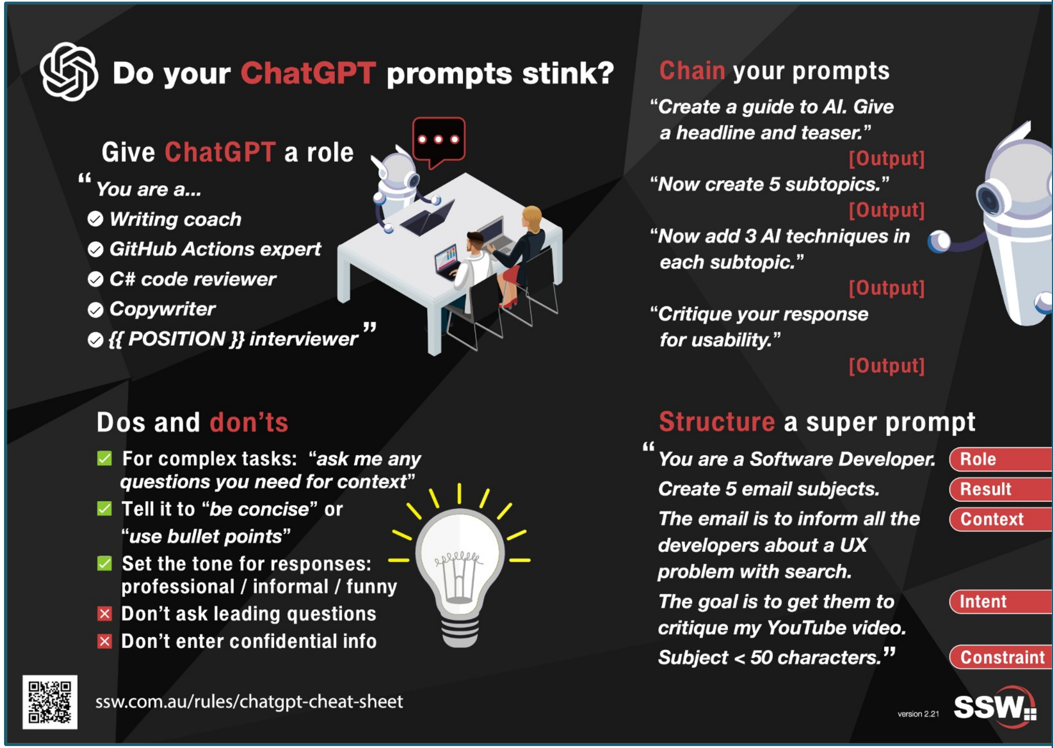
NAIRR Pilot National Artificial Intelligence Research Resource Pilot

AI Unlocked - Intro to LLM Coding

Prompt Engineering Primer

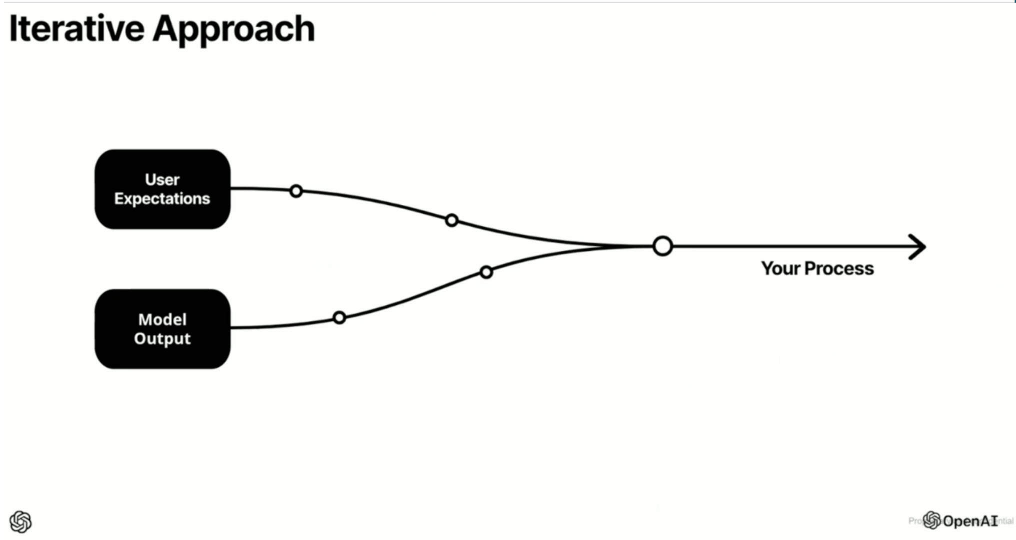
AI Unlocked - Intro to LLM Coding

- **Prompt engineering** is the process of designing and optimizing the language input to effectively guide a LLM response
- Requires **Context** and **Expectations** and can include a **Persona** (role or perspective)
- The more context and detailed expectations you provide the better the output.



AI Unlocked - Intro to LLM Coding

- **Prompt engineering** is usually an iterative process
- Can benefit from:
  - Delimiters can highlight logical breaks and tasks
  - Step-by-step instructions
  - One/few shot learning (give it examples of what you want)
  - Ask the LLM to help write an effective prompt!



https://academy.openai.com/public/videos/advanced-prompt-engineering-2025-02-13

Goals for the Tutorial

AI Unlocked - Intro to LLM Coding

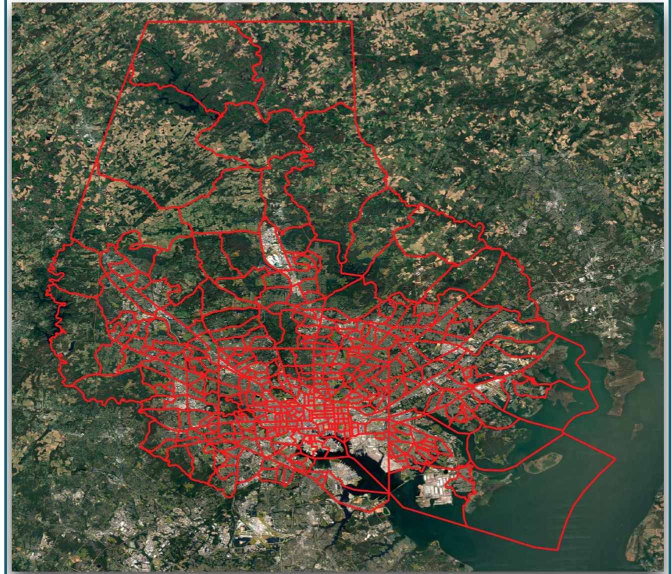
Goals for the Tutorial

- **Have FUN!!**
- Use large language models (LLMs) to generate and modify Python code in Google Colab
- Apply prompt-engineering strategies to iteratively debug code, resolve errors, and improve analytical workflows
- Integrate satellite remote sensing data (NDVI from NASA's Harmonized Landsat–Sentinel dataset) with U.S. Census socioeconomic datasets
- Summarize and visualize spatial patterns using maps, tables, and simple statistical comparisons
- Develop skills for using AI to write and refine code while creating a reproducible workflow that can be adapted to other geospatial datasets and research questions.

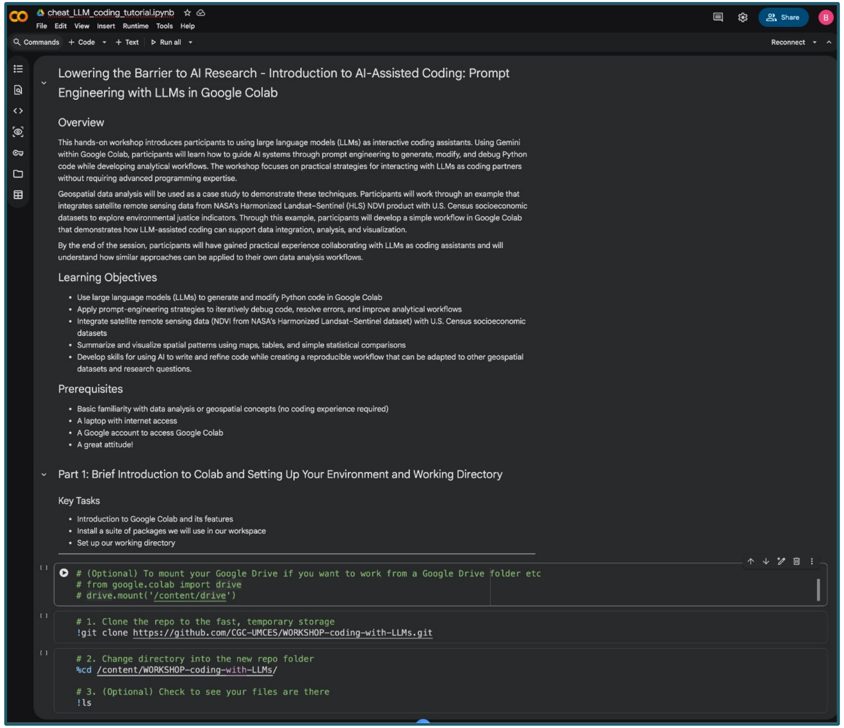
Visual Roadmap for the Tutorial

AI Unlocked - Intro to LLM Coding

Part 1: Introduction to Google Colab environment

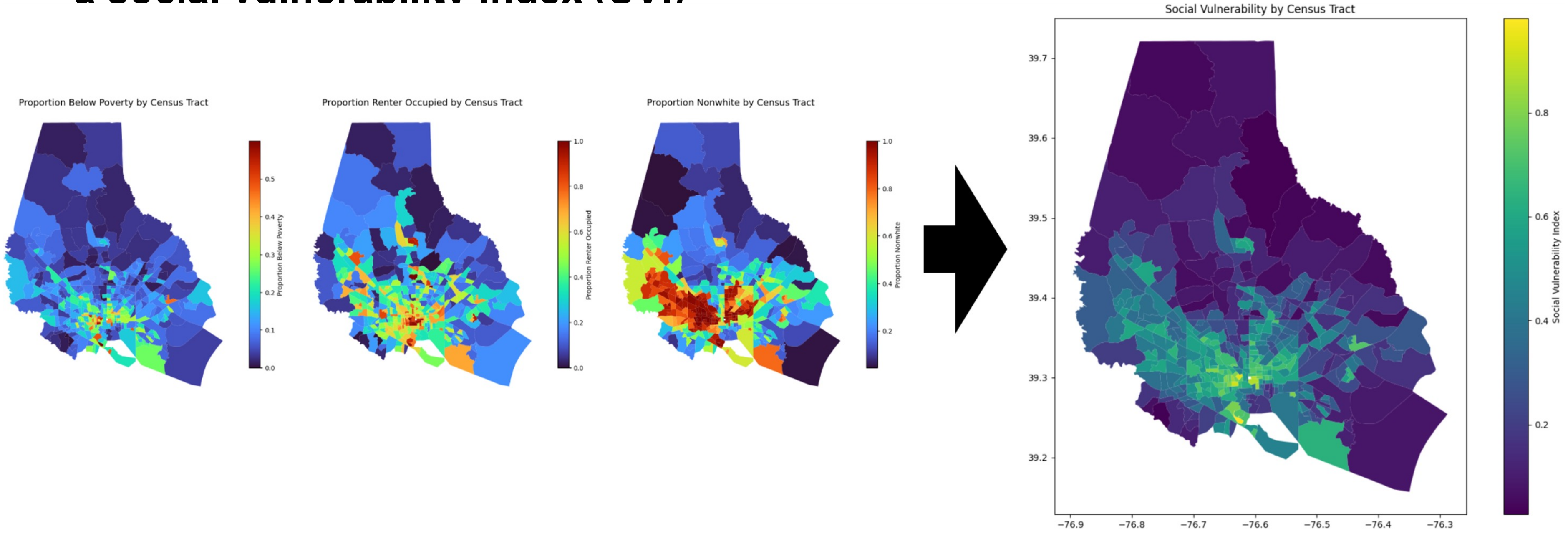


Baltimore City and County Census Tracts we will use



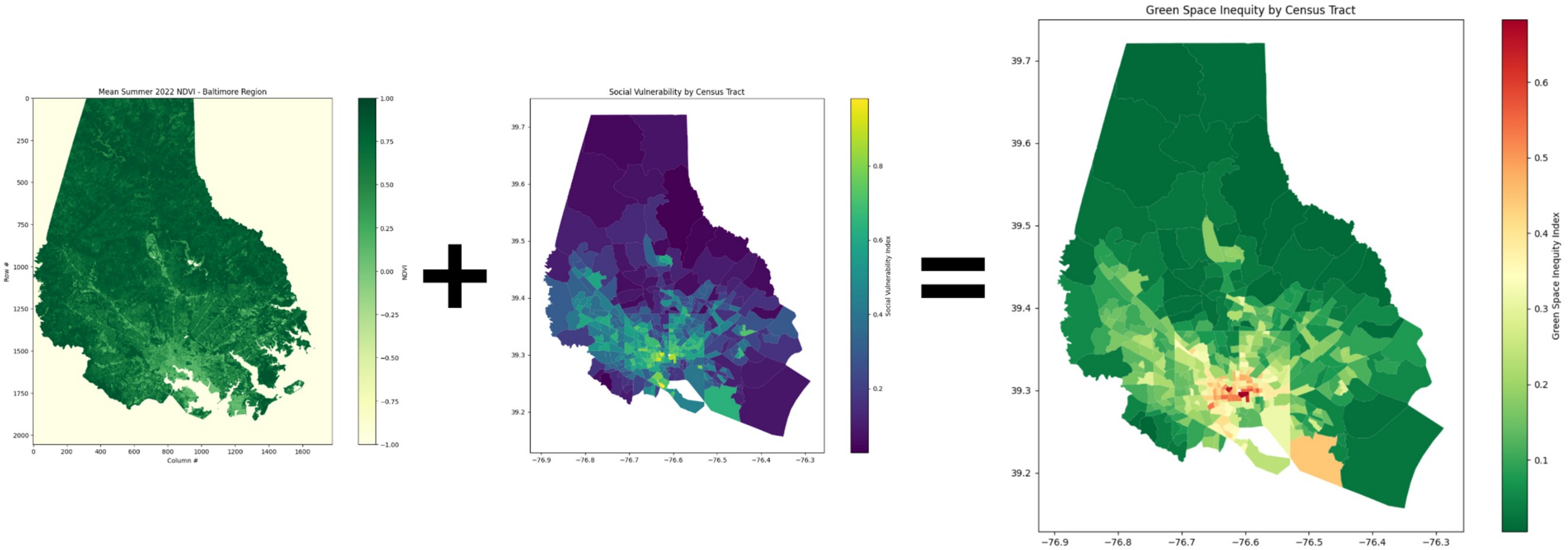
AI Unlocked - Intro to LLM Coding

Part 2: Working with Census data to create a social vulnerability index (SVI)



AI Unlocked - Intro to LLM Coding

Part 3: Integrating Harmonized Landsat-Sentinel 2 NDVI data with the SVI from Part 2 to build a Green Space Inequity Index

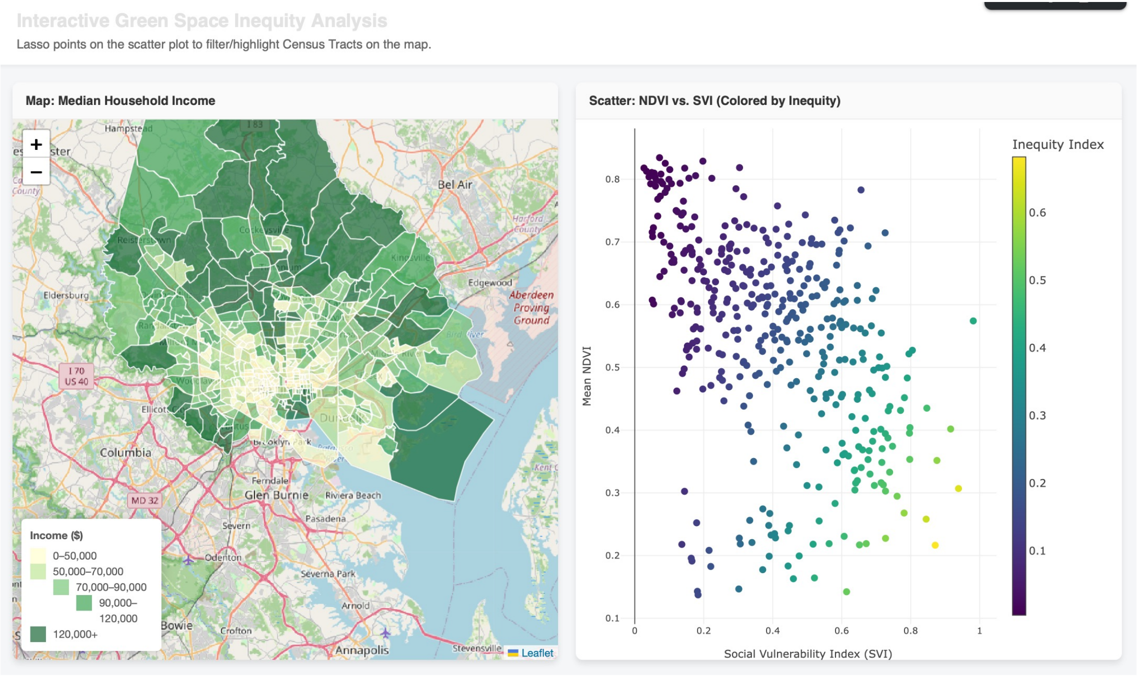


AI Unlocked - Intro to LLM Coding

Part 4: Freestyling with LLMs!

AI Unlocked - Intro to LLM Coding

Part 5: Creating a Shareable Interactive Data Dashboard



Research Computing
UNIVERSITY OF COLORADO BOULDER

ACCESS
Support

NAIRR Pilot
National Artificial Intelligence Research Resource Pilot

AI Unlocked - Intro to LLM Coding

Discussion Q & A (last ~5-10 mins)

- Discussion of the strengths and limitations of LLM-assisted analysis
- Best practices for validating AI-generated code and maintaining reproducibility
- Opportunities and challenges for AI in geospatial research
- Audience questions and discussion

AI Unlocked - Intro to LLM Coding

Thank You!

Questions?

Burch Fisher | burch.fisher@umces.edu

Xiaoxu Guo | xguo@umces.edu

University of Maryland
CENTER FOR ENVIRONMENTAL SCIENCE

Github Repository URL
https://github.com/CGC-UMCES/NAIRR-2026-Tutorial-Intro-Colab-LLM-Coding